ASSIGNMENT No:-05

Write a python Program for Bidirectional Associative Memory with two pairs of vectors.

In [1]:
```
import numpy as np
```

In [4]:
```
X=np.array([
    [1,1,-1,-1],
    [1,-1,1,-1]
])
Y=np.array([
    [1,1,-1],
    [1,-1,1]
])
```

In [5]:
```
print("Initial X Values\n",X)
print("Initial Y values\n",Y)
```

```
Initial X Values
 [[ 1  1 -1 -1]
 [ 1 -1  1 -1]]
Initial Y values
 [[ 1  1 -1]
 [ 1 -1  1]]
```

In [6]:
```
w=np.zeros((X.shape[1],Y.shape[1]))
```

In [8]:
```
def bipolar(x):
    return np.where(x>=0,1,-1)
```

In [11…
```
for i in range(len(X)):
    w+=np.outer(X[i],Y[i])
print("Final Weight Matrix\n",w)
```

```
Final Weight Matrix
 [[ 6.  0.  0.]
 [ 0.  6. -6.]
 [ 0. -6.  6.]
 [-6.  0.  0.]]
```

In [13…
```
print("Recall From X\n")
for x in X:
    y=bipolar(np.dot(x,w))
print(f"Input {x} -> Output {y}")
```

```
Recall From X

Input [ 1 -1  1 -1] -> Output [ 1 -1  1]
```

In [14…

```
print("Recall From Y\n")
for y in Y:
    x=bipolar(np.dot(y,w.T))
print(f"Input {y} -> Output {x}")
```

```
Recall From Y

Input [ 1 -1  1] -> Output [ 1 -1  1 -1]
```

In [ ]: